Homework 2

GENERAL ASSIGNMENT INSTRUCTIONS regarding the use of AI tools

AI tools like ChatGPT can be a powerful resource for generating ideas and refining your work. However, it is essential to know how to use these tools effectively and critically. Make sure to verify the relevance and accuracy of the answers you receive and be mindful of how AI-generated content aligns with the requirements of the assignment. Your own judgment and understanding are key to producing high-quality work.

Along with your responses, please append a list of any requests made to AI tools, such as ChatGPT, if used during the process of creating your answer.

Problem 1 (Elderly hospitalization): You are conducting a study on hospital utilization among elderly residents in a small Iowa town. The focus is on the number of days each individual spent as an inpatient at the local hospital during 2023. To collect data, you begin with a complete list of all elderly residents in the town and then randomly select eleven individuals. For each of these residents, you record the total number of inpatient hospital days in 2023. The data are: 12, 3, 0, 0, 0, 4, 6, 7, 32, 14, 10. The data are also provided, for your convenience, in the file ***hospital.csv***.

(a) Calculate and report the mean of these data.

(b) Calculate and report the median of these data.

(c) Calculate and report the standard deviation of these data.

Notes: You can calculate the mean, median, and standard deviation by hand or use R.

For the next two parts, assume that patterns of hospitalization in 2026 will be the same as those in 2023. Also ignore the potential increase in hospitalization with age.

(d) Imagine that your grandmother is an elderly resident of this small town. She hears about your work and asks you “how many days am I likely to be in hospitalized in 2026?” What number would you tell her? Briefly explain your choice.

(e) Imagine that the administrator of the only hospital in town asks you: “Can you estimate the total number of person-days that the elderly of this town will be in the hospital in 2026?” There are predicted to be 605 elderly residents of the town in 2026. What number would you tell her? Briefly explain your choice.

Note: If you aren't familiar with the concept of person-days, 1 person staying 5 days is 5 person-days. 5 people staying 1 day each is also 5 person-days. One staying 2 days and a second staying 3 days is also 5 person-days. The average of 2 and 3 is 2.5. The last number can be calculated as 2 people each staying the average of 2.5 days

Problem 2 (Radon levels): The file ***radon.txt*** contains measurements from a simple random sample of 42 owner-occupied homes in Ramsey County, Minnesota. Each number shows the level of radon gas (in pCi/L) found in that home. Radon is a naturally occurring radioactive gas that can build up indoors and may pose health risks, so studying its levels helps us understand potential exposure in the community.

a) Draw a box plot of the radon concentrations and say whether the distribution is symmetrical or skewed. Your answer is the plot and your description.

b) Calculate and report the average and median radon concentration. Include units in your answers.

c) Would you expect the average and median for these data to be similar? Briefly explain your answer.

Note: the question is not asking whether the two values are identical - when estimated from data, they almost never are identical. It is asking whether you should expect them to be close.

d) Calculate the standard error of the mean radon concentration of owner-occupied houses in Ramsey County.

Imagine you have taken a second simple random sample of radon in Ramsey County, MN, homes. There are 250 homes in this sample.

e) Would you expect the average radon concentration from that second sample to be close to the mean you calculated in question (1c)? Something larger? something smaller? Briefly explain your answer.

f) Would you expect the standard error of the mean for the 2nd sample to be close to the se you calculated in question **(1d)**? Something smaller? Something larger? Briefly explain your answer.

Problem 3 (Mutagen effect): Scientists wanted to study whether a certain mutagen affects cell growth. They began with a large "master" cell culture and haphazardly took 9 vials from it. Four of these vials were randomly assigned to the control group (**no** mutagen), while the other five vials were assigned to the treatment group, where cells were exposed to 80 mg/ml **dose** of the mutagen.

After giving the cells time to grow, the researchers examined them under a microscope. For each vial, they looked at 100 cells and counted how many showed microsatellite nuclei (tiny abnormalities that can signal mutations).

The data (which are available in ***mutagen.csv***) provide information about the treatment (no or dose) with the following values:

- no: 0 dose (control): 0, 0, 0, 0
- dose: 80 mg/ml dose: 2, 3, 5, 11, 20

Answer the following questions based on these data.

a) Design questions: What are the treatments in this study? What, if anything, are they randomly assigned to? Can causal conclusions be statistically justified? Why or why not?

The goal of this study is to test whether the mutagen has any effect. Our null hypothesis is that the mutagen makes no difference—that is, the mean number of microsatellite nuclei is the same in the treatment group (80 mg/ml dose) and the control group. We will answer this question by considering a **permutation test**. We will use the difference in means (dose of 80 mg/ml - control) as our test statistic. *With this definition, a positive value means that the treatment group (80 mg/ml) had, on average, more microsatellite nuclei than the control group.* The observed difference (calculated from the data above) is 8.2.

Because we had 9 vials (4 in the control group and 5 in the treatment group), there are a total of 126 possible ways to split the vials into groups of 4 and 5. For each possible split, we can compute the test statistic (difference in means). When we sort all 126 test statistics from smallest to largest, we get the values below. Use these values to answer the following two questions (*you may check your work using a computer, but please make sure you know how to compute the appropriate probabilities from the distribution.*)

-9.35 -8.9 -8 -8 -8 -8 -8 -7.1 -7.1 -7.1 -7.1 -6.65 -6.65 -6.65 -6.65 -5.75 -5.75 -5.75 -5.75 -5.75 -5.75 -5.3 -4.4 -4.4 -4.4 -4.4 -3.95 -3.95 -3.95 -3.95 -3.05 -3.05 -3.05 -3.05 -3.05 -3.05 -3.05 -3.05 -3.05 -3.05 -2.15 -2.15 -2.15 -2.15 -2.15 -2.15 -1.7 -1.7 -1.7 -1.7 -1.7 -1.7 -1.25 -0.8 -0.8 -0.8 -0.8 -0.35 -0.35 -0.35 -0.35 0.1 0.1 0.1 0.1 1 1 1 1 1 1 1 1 1 1 1.9 1.9 1.9 1.9 1.9 1.9 2.35 2.35 2.35 2.35 2.35 2.35 3.25 3.25 3.25 3.25 3.7 3.7 3.7 3.7 4.6 4.6 4.6 4.6 4.6 4.6 5.05 5.05 5.05 5.05 5.05 5.05 5.95 5.95 5.95 5.95 5.95 5.95 5.95 5.95 5.95 5.95 6.85 6.85 6.85 6.85 7.3 7.3 7.3 8.2

b) If the null hypothesis (no difference) is true, what is the probability of 8.2 or a more extreme **positive** value?
c) What is the **two-sided** *p*-value for the test of the null hypothesis of no difference?

The following is an exercise on using R to perform numerical calculations.

d) Use R to conduct a randomization test, using 1000 samples of the mean difference. What is the two-sided *p*-value calculated in R?
e) Why is your answer to part **d**) slightly different from than in part **c**)?
f) Write a short conclusion (1 or perhaps a few sentences) about the effect of this mutagen on the number of microsatellite nuclei, using the permutation test for your inference. ***Note:*** *It is also useful to report the estimated effect, in addition to the statistical inference. For additional examples, you may consider the "Summary of Statistical Analysis" paragraphs (usually labeled as Statistical Conclusions) in the Statistical Sleuth at the end of each case study*.